ELASTICSEARCH WORKSHOP

Category 5: Boolean Compound Queries

EXERCISES

Scenario

You are a data engineer for a fictional intelligence unit called UNIT-7.
Your unit intercepts radio and digital communications (SIGINT).
Each intercept record has a source, content, priority, and status.
You will use bool queries to combine multiple conditions when searching.
All call signs, frequencies, and content are completely fictional.

Key Concept: The bool Query

A bool query combines other queries together. It has four sections:

Section	SQL equivalent	Meaning
`must`	AND	Document MUST match. Affects the relevance score.
`filter`	AND (fast)	Document MUST match. Does NOT affect the score. Cached. Use for yes/no conditions.
`should`	OR	Document SHOULD match. Matching boosts the score. Not required unless it is the only clause.
`must_not`	AND NOT	Document MUST NOT match. Excludes documents. Does not affect the score.

Rule: must and filter both require a match. Use filter for keyword, range, and boolean conditions because it is faster. Use must when relevance scoring matters (text search).

SQL analogy:
must = AND (with scoring)
filter = AND (no scoring, faster)
should = OR
must_not = AND NOT

Dataset: SIGINT Intercept Log

Index name: 'intercepts' | 12 records

intercept_id	source	channel	priority	language	content_flag	signal_strength	intercepted_on	analyzed
INT-001	ALPHA-1	radio	HIGH	Arabic	weapons	87	2024-01-08	true
INT-002	BRAVO-3	digital	LOW	Farsi	logistics	62	2024-01-15	false
INT-003	ALPHA-1	radio	HIGH	Arabic	movement	91	2024-01-22	true
INT-004	DELTA-2	radio	MEDIUM	Turkish	logistics	55	2024-02-03	false
INT-005	BRAVO-3	digital	HIGH	Farsi	weapons	78	2024-02-10	true
INT-006	CHARLIE-4	radio	LOW	Arabic	personal	44	2024-02-17	false
INT-007	ALPHA-1	digital	HIGH	Arabic	weapons	95	2024-02-24	true
INT-008	DELTA-2	radio	MEDIUM	Turkish	movement	70	2024-03-02	true
INT-009	BRAVO-3	radio	LOW	Farsi	personal	38	2024-03-09	false
INT-010	CHARLIE-4	digital	HIGH	Arabic	movement	83	2024-03-14	true
INT-011	ALPHA-1	radio	MEDIUM	Arabic	logistics	66	2024-03-18	false
INT-012	DELTA-2	digital	HIGH	Farsi	weapons	90	2024-03-22	true

signal_strength is an integer from 1 to 100. Higher = stronger signal.

Exercise 1 [Step-by-step]

Task: Create the 'intercepts' index with the correct mapping. Bulk index all 12 records.

Step 1 — Create the index

```
from elasticsearch import Elasticsearch

es = Elasticsearch('http://localhost:9200')

if es.indices.exists(index='intercepts'):
    es.indices.delete(index='intercepts')

mapping = {
    'mappings': { 'properties': {
        'intercept_id':   { 'type': 'keyword' },
        'source':         { 'type': 'keyword' },
        'channel':        { 'type': 'keyword' },
        'priority':       { 'type': 'keyword' },
        'language':       { 'type': 'keyword' },
        'content_flag':   { 'type': 'keyword' },
        'signal_strength':{ 'type': 'integer' },
        'intercepted_on': { 'type': 'date'    },
```

```
        'analyzed':       { 'type': 'boolean' }
    }}
}
es.indices.create(index='intercepts', body=mapping)
print('Index created.')
```

Step 2 — Bulk index all 12 intercepts

```
from elasticsearch.helpers import bulk

records = [

{'_index':'intercepts','_id':'INT-001','_source':{'intercept_id':'INT-001','source'
:'ALPHA-1','channel':'radio','priority':'HIGH','language':'Arabic','content_flag':'
weapons','signal_strength':87,'intercepted_on':'2024-01-08','analyzed':True}},

{'_index':'intercepts','_id':'INT-002','_source':{'intercept_id':'INT-002','source'
:'BRAVO-3','channel':'digital','priority':'LOW','language':'Farsi','content_flag':'
logistics','signal_strength':62,'intercepted_on':'2024-01-15','analyzed':False}},

{'_index':'intercepts','_id':'INT-003','_source':{'intercept_id':'INT-003','source'
:'ALPHA-1','channel':'radio','priority':'HIGH','language':'Arabic','content_flag':'
movement','signal_strength':91,'intercepted_on':'2024-01-22','analyzed':True}},

{'_index':'intercepts','_id':'INT-004','_source':{'intercept_id':'INT-004','source'
:'DELTA-2','channel':'radio','priority':'MEDIUM','language':'Turkish','content_flag
':'logistics','signal_strength':55,'intercepted_on':'2024-02-03','analyzed':False}}
,

{'_index':'intercepts','_id':'INT-005','_source':{'intercept_id':'INT-005','source'
:'BRAVO-3','channel':'digital','priority':'HIGH','language':'Farsi','content_flag':
'weapons','signal_strength':78,'intercepted_on':'2024-02-10','analyzed':True}},

{'_index':'intercepts','_id':'INT-006','_source':{'intercept_id':'INT-006','source'
:'CHARLIE-4','channel':'radio','priority':'LOW','language':'Arabic','content_flag':
'personal','signal_strength':44,'intercepted_on':'2024-02-17','analyzed':False}},

{'_index':'intercepts','_id':'INT-007','_source':{'intercept_id':'INT-007','source'
:'ALPHA-1','channel':'digital','priority':'HIGH','language':'Arabic','content_flag'
:'weapons','signal_strength':95,'intercepted_on':'2024-02-24','analyzed':True}},

{'_index':'intercepts','_id':'INT-008','_source':{'intercept_id':'INT-008','source'
:'DELTA-2','channel':'radio','priority':'MEDIUM','language':'Turkish','content_flag
':'movement','signal_strength':70,'intercepted_on':'2024-03-02','analyzed':True}},

{'_index':'intercepts','_id':'INT-009','_source':{'intercept_id':'INT-009','source'
:'BRAVO-3','channel':'radio','priority':'LOW','language':'Farsi','content_flag':'pe
rsonal','signal_strength':38,'intercepted_on':'2024-03-09','analyzed':False}},

{'_index':'intercepts','_id':'INT-010','_source':{'intercept_id':'INT-010','source'
:'CHARLIE-4','channel':'digital','priority':'HIGH','language':'Arabic','content_fla
g':'movement','signal_strength':83,'intercepted_on':'2024-03-14','analyzed':True}},

{'_index':'intercepts','_id':'INT-011','_source':{'intercept_id':'INT-011','source'
:'ALPHA-1','channel':'radio','priority':'MEDIUM','language':'Arabic','content_flag'
:'logistics','signal_strength':66,'intercepted_on':'2024-03-18','analyzed':False}},

{'_index':'intercepts','_id':'INT-012','_source':{'intercept_id':'INT-012','source'
:'DELTA-2','channel':'digital','priority':'HIGH','language':'Farsi','content_flag':
'weapons','signal_strength':90,'intercepted_on':'2024-03-22','analyzed':True}},
```

```
]

success, errors = bulk(es, records)
print('Indexed:', success, '| Errors:', errors)
# Expected: Indexed: 12 | Errors: []
```

Exercise 2 [Step-by-step] (required)

Task: Find all intercepts where priority is HIGH AND channel is 'radio'.

Use a bool query with must. Both conditions must be true.
Print the intercept_id, source, priority, and channel of each result.

Step 1 — Write the bool must query

```
from elasticsearch import Elasticsearch

es = Elasticsearch('http://localhost:9200')

# bool query with must.
# must is a list. Every item in the list must match.
# This is AND logic: priority = HIGH AND channel = radio.
query = {
    'query': {
        'bool': {
            'must': [
                { 'term': { 'priority': 'HIGH'  } },
                { 'term': { 'channel':  'radio' } }
            ]
        }
    }
}
```

Step 2 — Run the search

```
response = es.search(index='intercepts', body=query)
hits = response['hits']['hits']

print('Results:', len(hits))
for hit in hits:
    src = hit['_source']
    print(src['intercept_id'], '-', src['source'],
          '- priority:', src['priority'], '- channel:', src['channel'])

# Expected output: 2 intercepts
# INT-001 - ALPHA-1 - priority: HIGH - channel: radio
# INT-003 - ALPHA-1 - priority: HIGH - channel: radio
```

```
Command reference:
  'bool': { 'must': [ condition1, condition2, ... ] }
  Every condition in must must be true. This is AND logic.
  Each condition can be any query: term, range, match, etc.

SQL analogy:     WHERE priority = 'HIGH' AND channel = 'radio'
MongoDB analogy: { priority: 'HIGH', channel: 'radio' }
```

Exercise 3

Task: Find all HIGH priority intercepts with signal_strength >= 80. Use filter, not must.

filter works like must. The difference is speed and scoring.
filter does not calculate a relevance score. It is faster for yes/no conditions.
Use filter for all keyword, range, and boolean conditions when scoring does not matter.

Print the intercept_id, priority, and signal_strength of each result.

Expected result: INT-001 (87), INT-003 (91), INT-007 (95), INT-010 (83), INT-012 (90). That is 5 intercepts.

```
Command reference:
  'bool': { 'filter': [ condition1, condition2 ] }

  filter = AND logic, same as must, but:
    - Results are NOT given a relevance score.
    - Filter results are cached by Elasticsearch. Repeated queries are faster.
    - Use filter for exact-match and range conditions.
    - Use must when you need text search scoring.

SQL analogy:     WHERE priority = 'HIGH' AND signal_strength >= 80
```

Exercise 4

Task: Find all intercepts where content_flag is 'weapons' OR content_flag is 'movement'.

Use a bool query with should. A document matches if it satisfies at least one condition.
Print the intercept_id, content_flag, and priority of each result.

Expected result: INT-001, INT-003, INT-005, INT-007, INT-008, INT-010, INT-012. That is 7 intercepts.

```
Command reference:
  'bool': { 'should': [ condition1, condition2 ] }

  should = OR logic. A document matches if ANY condition is true.
  When should is the only clause (no must or filter), at least one should must match.
  You can control this with minimum_should_match.

  'minimum_should_match': 1   At least 1 should condition must match (this is the default).
  'minimum_should_match': 2   At least 2 should conditions must match.

SQL analogy:     WHERE content_flag = 'weapons' OR content_flag = 'movement'
```

Exercise 5 (required)

Task: Find all intercepts that have NOT been analyzed yet (analyzed = false). Exclude LOW priority ones.

Use must_not to exclude documents that match a condition.
Condition 1 (filter): analyzed = false
Condition 2 (must_not): priority = 'LOW'

Print the intercept_id, priority, and analyzed of each result.

Expected result: INT-004 (MEDIUM), INT-011 (MEDIUM). That is 2 intercepts.

```
Command reference:
  'bool': {
      'filter':   [ { 'term': { 'analyzed': False } } ],
      'must_not': [ { 'term': { 'priority': 'LOW' } } ]
  }

  must_not excludes documents. A document that matches must_not is removed from results.
  must_not does not affect the score of remaining documents.

SQL analogy:     WHERE analyzed = false AND priority != 'LOW'
```

Exercise 6 (required)

Task: Find all intercepts where the language is 'Arabic', the priority is 'HIGH', AND the signal_strength is >= 85.

Use must for the text-relevant condition and filter for the exact-match and range conditions. This is the recommended Elasticsearch pattern: must for scoring, filter for everything else.

Print the intercept_id, language, priority, and signal_strength of each result.

Expected result: INT-003 (91), INT-007 (95). That is 2 intercepts.

```
Command reference:
  'bool': {
      'must':   [ { 'term': { 'language': 'Arabic' } } ],
      'filter': [
          { 'term':  { 'priority': 'HIGH' } },
          { 'range': { 'signal_strength': { 'gte': 85 } } }
      ]
  }

  Best practice: put scoring conditions in must, put yes/no conditions in filter.
  filter is cached and faster. It does not change which documents match,
  only whether the score is calculated for the filter conditions.

SQL analogy:     WHERE language = 'Arabic' AND priority = 'HIGH' AND signal_strength >= 85
```

Exercise 7

Task: Find all HIGH priority weapons intercepts that have NOT yet been analyzed.

Conditions:
- priority = HIGH (filter)
- content_flag = weapons (filter)
- analyzed = false (must_not the value true, OR filter for false)

Print the intercept_id, priority, content_flag, and analyzed of each result.

Expected result: INT-005. That is 1 intercept.

```
Command reference:
```

```
  'bool': {
      'filter':   [ { 'term': { 'priority':     'HIGH'    } },
                    { 'term': { 'content_flag': 'weapons' } } ],
      'must_not': [ { 'term': { 'analyzed': True } } ]
  }

SQL analogy:     WHERE priority = 'HIGH' AND content_flag = 'weapons' AND analyzed = false
```

Exercise 8

Task: Find intercepts from source ALPHA-1 OR BRAVO-3 OR DELTA-2 that are HIGH priority.

Use filter for the priority condition.
Use should for the three source options. Set minimum_should_match to 1.

Print the intercept_id, source, and priority of each result.

Expected result: INT-001, INT-003, INT-005, INT-007, INT-012. That is 5 intercepts.

```
Command reference:
  'bool': {
      'filter': [ { 'term': { 'priority': 'HIGH' } } ],
      'should': [
          { 'term': { 'source': 'ALPHA-1' } },
          { 'term': { 'source': 'BRAVO-3' } },
          { 'term': { 'source': 'DELTA-2' } }
      ],
      'minimum_should_match': 1
  }

  When filter or must is present alongside should, should conditions are optional
  unless you set minimum_should_match.
  Setting minimum_should_match: 1 forces at least one should to match.

SQL analogy:     WHERE priority = 'HIGH' AND (source = 'ALPHA-1' OR source = 'BRAVO-3' OR
source = 'DELTA-2')
```

Exercise 9

Task: Find intercepts that are either (HIGH priority AND weapons) OR (MEDIUM priority AND movement).

This requires a nested bool query. A should clause contains two inner bool queries.
Think of it as: (A AND B) OR (C AND D).
Each inner bool query handles one AND group. The outer should handles the OR.

Print the intercept_id, priority, and content_flag of each result.

Expected result: INT-001, INT-005, INT-007, INT-008, INT-012. That is 5 intercepts.

```
Command reference:
  'bool': {
      'should': [
          { 'bool': { 'filter': [ { 'term': { 'priority': 'HIGH' } },
                                  { 'term': { 'content_flag': 'weapons' } } ] } },
          { 'bool': { 'filter': [ { 'term': { 'priority': 'MEDIUM' } },
                                  { 'term': { 'content_flag': 'movement' } } ] } }
      ],
      'minimum_should_match': 1
  }

  A bool query can contain other bool queries. This is called nesting.
  Use nesting whenever you need (A AND B) OR (C AND D) logic.

SQL analogy:     WHERE (priority='HIGH' AND content_flag='weapons')
                 OR    (priority='MEDIUM' AND content_flag='movement')
```

Exercise 10

Task: Build a complex query. Find HIGH or MEDIUM priority intercepts in Arabic, with signal strength between 60 and 95, that have been analyzed, and are NOT about personal content.

Break it down:
- language = Arabic (filter)
- signal_strength between 60 and 95 (filter, range)
- analyzed = true (filter)
- content_flag != personal (must_not)
- priority = HIGH or MEDIUM (should, minimum_should_match: 1)

Print the intercept_id, priority, language, signal_strength, content_flag, and analyzed.

Expected result: INT-001 (HIGH, weapons, 87), INT-003 (HIGH, movement, 91), INT-008 (MEDIUM, movement, 70), INT-010 (HIGH, movement, 83). That is 4 intercepts.

```
Build this step by step:
  Start with an empty bool query.
  Add filter conditions one by one.
  Add must_not.
  Add should with minimum_should_match.
  Test after each addition.

SQL analogy:
  WHERE language = 'Arabic'
  AND signal_strength BETWEEN 60 AND 95
  AND analyzed = true
  AND content_flag != 'personal'
  AND (priority = 'HIGH' OR priority = 'MEDIUM')
```

End of Category 5 Exercises